MCTR911 | ROBOTICS PROGRAMMING

5-DOF Robot Arm
DH Kinematic Analysis

Forward Kinematics Technical Report

TEAM 06

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Scope

This report documents the supplied 5-DOF kinematic model. The team literature review describes a separate UR10e 6-DOF robot concept.

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Forward Kinematics of a Five-Degree-of-Freedom Robot Arm

Technical Report | Denavit–Hartenberg Method

Prepared for academic submission

1. Objective

Determine the forward kinematic model of a five-degree-of-freedom serial robot manipulator using the Denavit–Hartenberg (DH) convention. The model relates the five joint angles to the position and orientation of the end-effector.

2. Robot configuration

The manipulator consists of five revolute joints with joint variables $\theta1$, $\theta2$, $\theta3$, $\theta4$ and $\theta5$. The link coordinate frames follow the arrangement in the supplied hand-drawn robot sketch. The DH parameters below reproduce the parameter structure in the source.

3. Denavit–Hartenberg parameters

Joint i	θi	di	ai	αi
1	θ1	L1	0	–π/2
2	θ2	0	L2	0
3	θ3	0	L3	0
4	θ4	0	0	π/2
5	θ5	L5	0	0

The source lists physical link lengths in metres; because some handwritten digits are not sufficiently legible, the derivation retains the symbolic lengths L1, L2, L3 and L5. These should be checked against the original measurement sheet before numerical evaluation.

4. General homogeneous transformation

Using the standard DH convention, each consecutive frame transformation is:

$A_i = \mathrm{Rot}(z, \theta_i)\,\mathrm{Trans}(z, d_i)\,\mathrm{Trans}(x, a_i)\,\mathrm{Rot}(x, \alpha_i)$.

With $c_i = \cos(\theta_i)$ and $s_i = \sin(\theta_i)$, the standard matrix is:

$A_i = [\ c_i,\ -s_i\cos\alpha_i,\ s_i\sin\alpha_i,\ a_i c_i\ ;\ s_i,\ c_i\cos\alpha_i,\ -c_i\sin\alpha_i,\ a_i s_i\ ;\ 0,\ \sin\alpha_i,\ \cos\alpha_i,\ d_i\ ;\ 0,\ 0,\ 0,\ 1\ ]$.

5. Individual joint transformations

Each matrix below is obtained by substituting its row of the DH table into the standard transformation formula. Matrix rows are separated by semicolons.

A1 = [c1, 0, –s1, 0 ; s1, 0, c1, 0 ; 0, –1, 0, L1 ; 0, 0, 0, 1]

A2 = [c2, –s2, 0, L2 c2 ; s2, c2, 0, L2 s2 ; 0, 0, 1, 0 ; 0, 0, 0, 1]

A3 = [c3, –s3, 0, L3 c3 ; s3, c3, 0, L3 s3 ; 0, 0, 1, 0 ; 0, 0, 0, 1]

A4 = [c4, 0, s4, 0 ; s4, 0, –c4, 0 ; 0, 1, 0, 0 ; 0, 0, 0, 1]

A5 = [c5, –s5, 0, 0 ; s5, c5, 0, 0 ; 0, 0, 1, L5 ; 0, 0, 0, 1]

6. Overall forward kinematics

The base-to-end-effector homogeneous transformation is obtained by ordered multiplication:

$T_{0,5}$ = A1 A2 A3 A4 A5.

Define θ23 = θ2 + θ3, θ234 = θ2 + θ3 + θ4, c23 = cos(θ23), s23 = sin(θ23), c234 = cos(θ234), s234 = sin(θ234). The final rotation matrix R and position vector p satisfy $T_{0,5}$ = [R | p; 0 0 0 | 1].

Position:

p_x = c1 (L2 c2 + L3 c23 + L5 s234)

p_y = s1 (L2 c2 + L3 c23 + L5 s234)

p_z = L1 – L2 s2 – L3 s23 + L5 c234

Orientation (rotation matrix rows):

R1 = [c1 c234 c5 – s1 s5, –c1 c234 s5 – s1 c5, c1 s234]

R2 = [s1 c234 c5 + c1 s5, –s1 c234 s5 + c1 c5, s1 s234]

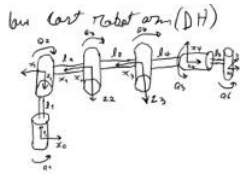
R3 = [–s234 c5, s234 s5, c234]

7. Conclusion

The forward kinematics of the five-revolute-joint manipulator has been established symbolically from its DH parameters. Once the link lengths and joint angles are supplied, the transformation gives the end-effector Cartesian position and orientation relative to the base frame. The source image is reproduced on the next page for traceability.

Appendix A — Original handwritten derivation

Original sketch, parameter table and handwritten intermediate transformations supplied for this report.



Joint (i)	θ_i	d_i	a_i	α_i
1	q_1	l_1	0	$-\pi/2$
2	q_2	0	l_2	0
3	q_3	0	l_3	0
4	q_4	0	0	$\pi/2$
5	q_5	l_5	0	0

From Solidworks:

$L_1 = 0{,}0471\ m$

$L_2 = 0{,}110\ m$

$L_3 = 0{,}107\ m$

$L_4 = 0{,}047\ m$

Transformation matrices:

$$^0T_1 = \begin{bmatrix} c_{q_1} & 0 & -s_{q_1} & 0 \\ s_{q_1} & 0 & c_{q_1} & 0 \\ 0 & -1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^1T_2 = \begin{bmatrix} c_{q_2} & -s_{q_2} & 0 & l_2 c_{q_2} \\ s_{q_2} & c_{q_2} & 0 & l_2 s_{q_2} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^2T_3 = \begin{bmatrix} c_{q_3} & -s_{q_3} & 0 & l_3 c_{q_3} \\ s_{q_3} & c_{q_3} & 0 & l_3 s_{q_3} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^3T_4 = \begin{bmatrix} c_{q_4} & 0 & s_{q_4} & 0 \\ s_{q_4} & 0 & -c_{q_4} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^4T_5 = \begin{bmatrix} c_{q_5} & -s_{q_5} & 0 & 0 \\ s_{q_5} & c_{q_5} & 0 & 0 \\ 0 & 0 & 1 & l_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{EE} = {}^0T_1 * {}^1T_2 * {}^2T_3 * {}^3T_4 * {}^4T_5$$

$$^0T_2 = \begin{bmatrix} c_{q_1}c_{q_2} & -c_{q_1}s_{q_2} & -s_{q_1} & c_{q_1} l_2 c_{q_2} \\ s_{q_1}c_{q_2} & -s_{q_1}s_{q_2} & c_{q_1} & s_{q_1} l_2 c_{q_2} \\ -s_{q_2} & -c_{q_2} & 0 & l_1 - l_2 s_{q_2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^0T_3 = \begin{bmatrix} c_{q_1 q_2 q_3} & -c_{q_1 q_2 q_3} & -s_{q_1} & c_{q_1}(l_2 c_{q_2} + l_3 c_{q_2 q_3}) \\ s_{q_1 q_2 q_3} & -s_{q_1 q_2 q_3} & c_{q_1} & s_{q_1}(l_2 c_{q_2} + l_3 c_{q_2 q_3}) \\ -s_{q_2 q_3} & -c_{q_2 q_3} & 0 & l_1 - l_2 s_{q_2} - l_3 s_{q_2 q_3} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^0T_4 = \begin{bmatrix} c_{q_1 q_2 q_3 q_4} & -s_{q_1} & c_{q_1 q_2 q_3} & c_{q_1}(l_2 c_{q_2} + l_3 c_{q_2 q_3}) \\ s_{q_1 q_2 q_3 q_4} & c_{q_1} & s_{q_1 q_2 q_3} & s_{q_1}(l_2 c_{q_2} + l_3 c_{q_2 q_3}) \\ -s_{q_2 q_3 q_4} & 0 & c_{q_2 q_3 q_4} & l_1 - l_2 s_{q_2} - l_3 s_{q_2 q_3} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^0T_5 = \begin{bmatrix} c_{q_1 q_2 q_3 q_4} c_{q_5} - s_{q_1} s_{q_5} & -c_{q_1 q_2 q_3 q_4} s_{q_5} - s_{q_1} c_{q_5} & c_{q_1 q_2 q_3} & c_{q_1}(l_2 c_{q_2} + l_3 c_{q_2 q_3} + l_4 c_{q_2 q_3 q_4}) \\ s_{q_1 q_2 q_3 q_4} c_{q_5} + c_{q_1} s_{q_5} & -s_{q_1} c_{q_2 q_3 q_4} s_{q_5} + c_{q_1} c_{q_5} & s_{q_1 q_2 q_3} & s_{q_1}(l_2 c_{q_2} + l_3 c_{q_2 q_3} + l_4 c_{q_2 q_3 q_4}) \\ -s_{q_2 q_3 q_4} c_{q_5} & s_{q_2 q_3 q_4} s_{q_5} & c_{q_2 q_3 q_4} & l_1 - l_2 s_{q_2} - l_3 s_{q_2 q_3} - l_4 s_{q_2 q_3 q_4} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$